Lesson 5.1 - Introduction to Probability

> **Video:** Introduction to Probability, Basic Overview - Sample Space, & Tree Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Probability of an event

$$P(A) = \frac{\text{number of favorable outcomes (outcomes in } A)}{\text{total number of possible outcomes}}$$

This counting formula works when all outcomes are **equally likely**.

Range of values

$$0 \leq P(A) \leq 1$$

$P(A)$	Meaning
0	Impossible, it never happens
1	Certain, it always happens (100%)
0.3	30% chance: about 3 times in 10 tries, or 30 in 100

Example: if 20% of a city's residents drive a blue car, then picking 100 people at random should give **about 20** blue-car drivers, and picking 1,000 should give about 200.

Sample space and tree diagrams

The **sample space** S is the set of **all possible outcomes**.

One coin: $S = \{H, T\}$

Two coins: use a tree diagram.

```
        H — HH
    H <
        T — HT
start
        H — TH
    T <
        T — TT
```

$S = \{HH, HT, TH, TT\}$, so 4 outcomes.

Three coins:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, so $2^3 = 8$ outcomes.

> In general, flipping n coins gives 2^n equally likely outcomes.

Example 1 - coins

(a) Two coins: P(at least one head)? Favorable: $\{HH, HT, TH\}$, which is 3 of 4 outcomes.

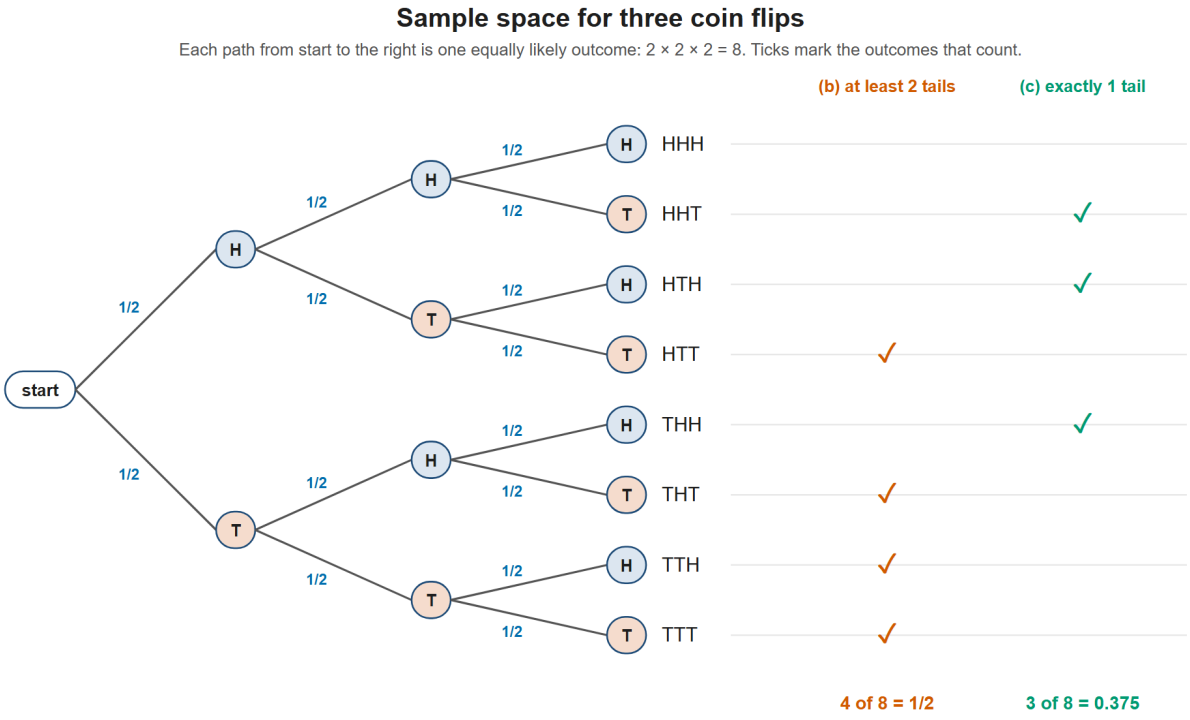
$$P = \frac{\text{favorable outcomes}}{\text{total outcomes}} = \frac{3}{4} = 0.75 = 75\%$$

(b) Three coins: P(at least two tails)? Favorable: $\{HTT, THT, TTH, TTT\}$, which is 4 of 8.

$$P = \frac{\text{favorable outcomes}}{\text{total outcomes}} = \frac{4}{8} = \frac{1}{2} = 50\%$$

(c) Three coins: P(exactly one tail)? Favorable: $\{HHT, HTH, THH\}$, which is 3 of 8.

$$P = \frac{\text{favorable outcomes}}{\text{total outcomes}} = \frac{3}{8} = 0.375 = 37.5\%$$

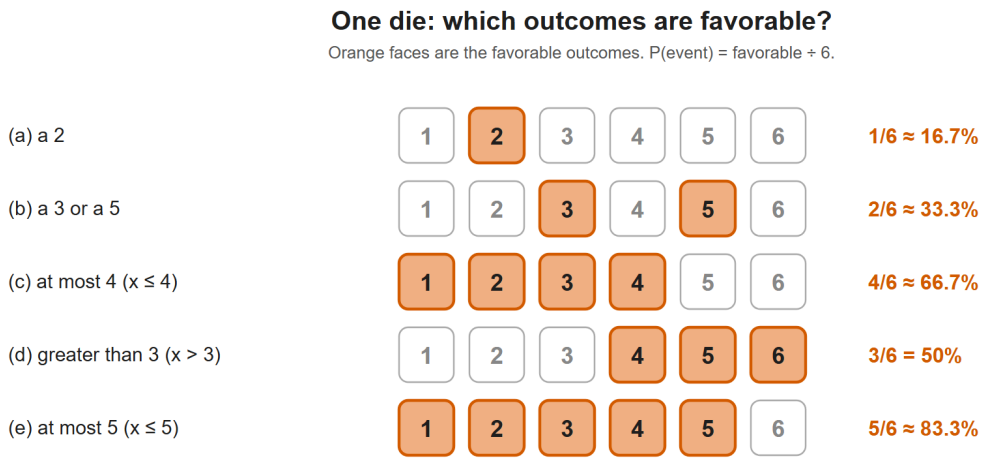


Every path through the tree is one of the 8 equally likely outcomes. Count the ticks: 4 outcomes have at least two tails (4/8 = 1/2), and 3 have exactly one tail (3/8).

Example 2 - one six-sided die

$S = \{1, 2, 3, 4, 5, 6\}$

Event	Favorable outcomes	Probability $= \frac{\text{favorable}}{6}$
(a) a 2	{2}	$\frac{1}{6} \approx 0.167$ (16.7%)
(b) a 3 or a 5	{3, 5}	$\frac{2}{6} = \frac{1}{3} \approx 33.3\%$
(c) **at most** 4 ($x \leq 4$)	{1, 2, 3, 4}	$\frac{4}{6} = \frac{2}{3} \approx 66.7\%$
(d) **greater than** 3 ($x > 3$, so 3 is excluded)	{4, 5, 6}	$\frac{3}{6} = \frac{1}{2} = 50\%$
(e) at most 5 ($x \leq 5$)	{1, 2, 3, 4, 5}	$\frac{5}{6} \approx 83.3\%$



The favorable faces for each die question. Note how "at most 4" includes 4, while "greater than 3" leaves 3 out.

Key takeaways

- $P(A)$ = favorable ÷ total, when the outcomes are equally likely.
- Every probability is between 0 and 1. Multiply by 100 for a percentage.
- List the **sample space**; a **tree diagram** helps for multi-step experiments.
- Read the wording carefully: "at least", "at most", "greater than" (strict), "exactly".